Potential Campsite Locations on Vancouver Island

GEOS 270 Final Project Report

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2024/12/14

(Link to Story Map)

I. Introduction

As Vancouver Island's population grows and BC as a whole, so does the demand for recreational opportunities like camping. To ensure continued access to outdoor activities, our group aims to identify optimal sites for new campgrounds. Various datasets were analyzed to create a filter for site selection. Key considerations include: proximity to hiking trails, distance from existing campgrounds, access to freshwater sources (rivers, lakes), elevation profiles, accessibility from roads as well as fire risk.

By applying limits to these criteria, promising locations on Vancouver Island were systematically identified. Some of these thresholds were set by the Government of British Columbia and others were determined by our group based on specific needs and desirable traits for campgrounds. The resulting analysis will highlight areas that contain all of the necessary requirements of accessibility, safety, and environmental impact, providing ideal locations for the development of new campgrounds.

II. Methodology

Data acquisition

In order to identify potential campsite locations, we listed factors of consideration, which include planning within provincial land, proximity to trails and water resources, slope of sites, flood risks, ecological sensitivity, noise level, and proximity to existing sites. We refined our list by inquiring about campsite planning conventions on BC Parks. Much of the data for this was available for download through the BC Data Catalogue. The digital elevation map was accessed through the USGM while the trail data was accessed by querying OpenStreetMaps using the QuickOSM tool in QGIS and querying for any object keys matching “highway:path” or “highway:footpath” in Vancouver Island. To minimize the potential sources of error, we took care to select up-to-date data relevant to the analysis of our factors of consideration.

Data analysis and visualization

First, we chose NAD 1983 BC Environment Albers as the projection for our maps and ensured all layers followed this projection. This projection covering the entire province is used in most official maps of BC, and it is appropriate for our map as well since Vancouver Island extends across 2 UTM zones, excluding UTM projection from suitable projections. We then created a project boundary for Vancouver Island by creating a new dissolved layer of Victoria and Nanaimo Districts from the BC Regional Districts layer. After all layers are clipped to the project boundary to reduce file size, further analyses can be performed on different layers.

For the North and South Island DEM files, the mosaic to new raster tool is used to combine them. The resulting raster layer is first reprojected and then used to produce slope percentages with the slope tool. It is further reclassified to match the 5 categories on the BC slope guidelines from the Recreation Manual. According to Chapter 9 of the guide, slopes between 2% and 5% are “most suitable for road, campsite and building development” and slopes between 5% and 10% have “marginal suitability for campsite and building development” (British Columbia

Ministry of Forests, 1991, p.17). These 2 categories are class 1 and 2 respectively. The reclassified raster layer is converted to a polygon, where the areas of class 1 and 2 are extracted as a new feature displaying areas that fulfill the category of slope for campsites. By creating a multiple-ring buffer of 20m to 100m for the trail layer, areas that are close to existing trails could be identified while maintaining some privacy for the campsites.

To identify areas close to freshwater sources, layers of lakes and rivers are combined through the union tool. According to Chapter 9 of the Recreation Manual of BC, a suitable potable water source within 400 meters would be required for walk-in tenting areas (British Columbia Ministry of Forests, 1991, p.49). Since the area of potential sites is not covered in the guide, an extra 100 meters for campsite space is added to our buffers. To avoid flood risks and environmental disturbance, we ensured campsites were not located within 20 meters of water bodies. According to the Flood Hazard Area Land Use Management Guidelines of BC, structures should be set back at least 15 meters from the natural boundaries of lakes (Ministry of Water, Land and Air Protection, 2004, p.16) – considering that campers are not protected by flood-proof infrastructures, we used 20 meters as the parameter instead for extra safety. Using the multiple-ring buffer tool, buffer areas of 20 m to 500 m are produced from the freshwater polygons. Using the intersect tool, areas that fulfill our criteria for potential sites from analysis of slope, trails, and water proximity are combined as a new layer.

Next, we analyzed the rest of the downloaded data to identify areas that should be eliminated from potential sites. Considering that campsites should be away from populated regions and noise pollution, we used paved roads as a proxy to eliminate unsuitable areas. Since we aim to identify campsite locations for backpackers, it is sufficient for sites to be only connected to trails instead of paved roads. From the Digital Road Atlas dataset, paved roads are selected to produce a new layer. To identify backcountry sites that would provide campers with an immersive experience, a buffer of 3 km is applied to paved roads. While BC Parks (n.d.) identifies areas more than 1 km away from highway or park roads as backcountry, we want to ensure there is minimum noise and air pollution, which may be generated from front-country campsites that allow vehicles as well.

Coastal regions on Vancouver Island are prone to flooding, as well as tsunamis. On the Regional District of Nanaimo's website, the tsunami hazard map for Vancouver Island shows that vulnerable areas on the west coast are at risk of waves up to 15 m (n.d.). In 1694, Vancouver Island was struck by a tsunami caused by the Alaska Earthquake, where the second wave reaching 4.3 m above the geodetic datum created an impact on infrastructures 1 km inland (Clague et al., 2003). Considering that there is a risk of even larger waves as suggested by the Government of BC, we set buffers for coastlines to be 2km. Taking into account the risk of flooding in low-lying areas and the continuous rise of sea levels, we decided to eliminate elevations below 25 m near the coastline. Using the DEM data that had been converted to slope percentages by the slope tool, we reclassified the layer into 2 classes: 0 - 25 m and greater than 25 m. The reclassified layer is converted with the raster to polygon tool, where the 0 to 25 m elevations, classified as class 0, are extracted as a new layer to intersect with the buffered coastline layer. The new layer produced with the intersect tool represents the areas that are at high risk of coastal flooding.

Historical Fire Perimeter Data is used to identify ecologically sensitive areas that were impacted by wildfires. Using the potential to reach canopy cover of over 10% and a minimum height of 5 m as the benchmarks for recovery, research results of ecosystem monitoring in Canada have shown that benchmarks were achieved on an average of 5 to 10 years (Bartels et al., 2016). Referring to the data above, we chose 10 years as the parameter to allow the recovery of wildlife and burnt environments. Selected polygons that represent areas with fire recorded from 2014 to 2024 are extracted as a new layer, showing areas to be eliminated.

We also excluded known campsites from our potential sites. A buffer of 2 km is applied, and the dissolve tool is used to create a new layer without overlapping buffer zones. The resulting layer is then unioned with the other layers of analysis results – buffers of paved roads, coastal flooding areas, and areas with fire records in the past 10 years. The unioned layer represents non-permitted areas that are to be removed.

Using the results from the analysis of areas that fulfill our criteria for potential sites, we erased the layer of non-permitted areas from the unioned layer of areas for potential sites. To ensure new sites are developed on provincial land instead of private properties, the potential campsite layer is clipped to the boundaries of BC Parks. The new clipped layer is then dissolved to remove overlapping areas. One criterion for new sites is the minimum size of 2000 square meters, hence we only selected areas fulfilling this criteria from the dissolved layer. The selected areas are extracted as a new layer, where distance from existing campsites is calculated using the near analysis tool. To visualize potential campsites in a clear and direct manner, we decided that points should be used as representations instead of polygons. Using the feature-to-point function, points are generated from the center of polygons in the final layer produced above.

III. Discussion

Data Collection and Analysis

To identify suitable campsite locations, several datasets were used, each offering important information. For example, wildfire data from the BC Wildfire Service informed natural hazard considerations, and Digital Elevation Models (DEMs) from GeoBC were used to evaluate slope suitability based on BC guidelines for the construction of recreation sites. Trail data from OpenStreetMaps and water resource data (rivers and lakes) from GeoBC ensured that recreational and practical needs were considered. Using the NAD 1983 BC Environment Albers projection ensured spatial accuracy, especially since Vancouver Island covers multiple UTM zones.

Some level of ground truthing is required to identify which shortlisted sites possess additional positive attributes that make them appealing to visitors. While many sites may meet basic suitability criteria, factors such as the presence of notable trails, scenic views, and nearby lakes or other attractions play a critical role in determining a site's desirability. Additionally, development must avoid creating visual disturbances to maintain visual quality objectives, especially from recreational water bodies like lakes and rivers. Curved access roads should be considered for drive-in sites to maintain privacy. Water bodies near potential sites must also be evaluated to determine their suitability as potable water sources, recreational features, or potential hazards, with appropriate testing or signage where necessary.

Limitations and Uncertainty

Despite efforts to ensure accuracy, some limitations were unavoidable. Wildfire data did not account for recent changes, such as vegetation recovery or fire prevention efforts, creating a temporal bias in the analysis. DEM resolution was enough for general terrain analysis but will have missed smaller, important details. These include features like gullies, minor elevation changes, or uneven ground, as well as localized hazards such as small cliffs, unstable ground, or steep drop-offs that might have been smoothed over or missed altogether.

Simplifications in the analysis also introduced errors. For example, converting slope data to polygons smoothed out details and misrepresented boundaries. Buffers for trails and water bodies assumed straight-line access, ignoring barriers such as dense forests. These kinds of simplifications could make some sites appear more suitable or unsuitable than they truly are. Field validation and better datasets can help address these issues. Additional hazards such as steep drop-offs, cliffs, avalanche areas, and other animal habitats must also be considered during site evaluations as not all of this data was readily available to the public. Campsite size requirements, like the minimum pad size of 8m diameter, and restrictions on structures in backcountry areas further limit site usability.

Suggestions for Improvement

Future research could make use of a more detailed, higher-resolution elevation raster, and incorporate additional datasets to improve accuracy. Visiting potential sites in person could validate the analysis results. Following this, in-person field tests in the proposed areas would need to be conducted to provide a cost estimate and environmental impact assessment. Finally, involving local communities, First Nations groups, and park officials would help ensure the results align with cultural, recreational, and conservation needs.

Conclusion

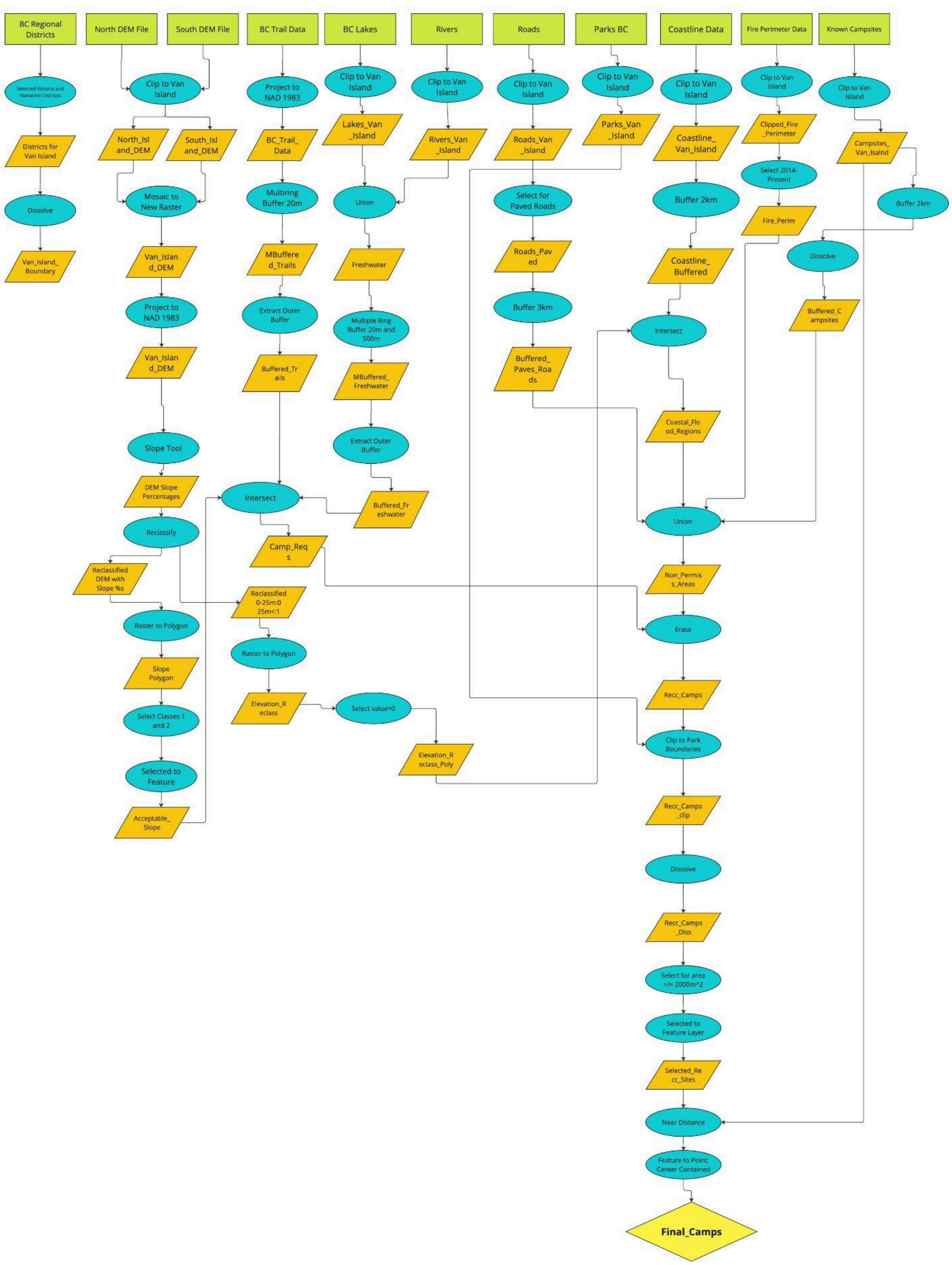
By using a variety of datasets, this analysis provided valuable insights into suitable campsite locations on Vancouver Island. However, addressing the data and methodological limitations will make future campsite planning more accurate and effective. This approach supports sustainable development while protecting the environment and meeting recreational needs.

IV. Dataset table

Table 1: Original Datasets

Datafile name	Source	Uses	Data model	Relevant attributes
Historical_ wildfire_perimeters	BC Wildfire Service, Nov. 27, 2024	Used to identify past wildfire locations to assess fire risks for potential campground sites	Vector	location, date, fire_type
Coastal_campgrounds	GeoBC Branch, Nov. 10, 2024	Used to identify existing coastal campgrounds and avoid clustering when selecting new campground locations.	Vector	location
Access_roads	GeoBC Branch, Nov. 10, 2024	Used to ensure accessibility to proposed campground sites, including proximity to existing roads	Vector	location
Rivers	GeoBC Branch, Nov. 10, 2024	Used to identify freshwater rivers near potential campgrounds for water access	Vector	location
Park_reserve_protected	BC Parks, Nov. 10, 2024	Used to find locations that the government has jurisdiction over and could use for a site.	Vector	location
Trails	Openstreetmap, Nov 27, 2024	Used to consider trail networks for recreational access near potential campground sites	Vector	location
DEM	GeoBC Branch Nov 27, 2024	Used to analyze terrain elevation and identify areas that are suitable for campground locations based on topography	Tabular -> XY table to point	elevation
Lakes	GeoBC Branch Nov 27, 2024	Identifies the location of lakes that may be relevant for water access or scenic value in potential campground locations	Vector	location
BC_regional_districts	Policy and Legislation	Used to clip all data	Polygon	county_names, locations
Coastline	GeoBC Branch Dec 7, 2024	Used to determine locations subject to coastal flooding	Vector	location

V. Flowchart



BC Regional Districts
Selected Victoria and Nanaimo Districts
Districts for Van Island
Dissolve
Van_Island_ Boundary
North DEM File
South DEM File
Clip to Van Island
North_Isl and_DEM
South_Isl and_DEM
Mosaic to New Raster
Van_Islan d_DEM
Project to NAD 1983
Van_Islan d_DEM
Slope Tool
DEM Slope Percentages
Reclassify
Reclassified DEM with Slope %s
Raster to Polygon
Slope Polygon
Select Classes 1 and 2
Selected to Feature
Acceptable_ Slope
BC Trail Data
Project to NAD 1983
BC_Trail_ Data
Multiring Buffer 20m
MBuffere d_Trails
Extract Outer Buffer
Buffered_Tr ails
Intersect
Camp_Req s
Reclassified 0-25m:0 25m<:1
Raster to Polygon
Elevation_R eclass
Select value=0
Elevation_R eclass_Poly
BC Lakes
Clip to Van Island
Lakes_Van _Island
Rivers
Clip to Van Island
Rivers_Van _Island
Union
Freshwater
Multiple Ring Buffer 20m and 500m
MBuffered_ Freshwater
Extract Outer Buffer
Buffered_Fr eshwater
Roads
Clip to Van Island
Roads_Van _Island
Select for Paved Roads
Roads_Pav ed
Buffer 3km
Buffered_ Paves_Roa ds
Parks BC
Clip to Van Island
Parks_Van _Island
Coastline Data
Clip to Van Island
Coastline_ Van_Island
Buffer 2km
Coastline_ Buffered
Intersect
Coastal_Flo od_Regions
Fire Perimeter Data
Clip to Van Island
Clipped_Fire _Perimeter
Select 2014- Present
Fire_Perim
Known Campsites
Clip to Van Island
Campsites_ Van_Isalnd
Buffer 2km
Dissolve
Buffered_C ampsites
Union
Non_Permis s_Areas
Erase
Recc_Camps
Clip to Park Boundaries
Recc_Camps _clip
Dissolve
Recc_Camps _Diss
Select for area >/= 2000m^2
Selected to Feature Layer
Selected_Re cc_Sites
Near Distance
Feature to Point; Center Contained
Final_Camps

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